

ADAI course portfolio

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Individual repository:

<https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025>

Group repository:

<https://github.com/FontysVenlo/groupepository-group-2>

Learning goal I

Proficiency level: Beginning

I consider myself a beginner in machine learning and neural networks, with limited theoretical knowledge and practical experience. My initial experience involved developing a basic image labeling AI model, which introduced me to concepts of artificial neural networks (ANNs) and convolutional neural networks (CNNs) in a practical context. (<https://github.com/Jazziki17/A-systems/tree/main/src>)

Planning:

- Week 1: research some information about CNNs as well as other topics discussed with the group. (<https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/Resources/Research%20questions%20and%20answers.txt>)
- Week 2: build a simple ai using DDI (diverse dermatology images) using tensorflow. (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions/aimodel.ipynb)
- Week 3: try again to build a CNN this time starting from a pretrained model. (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions_V2/model.ipynb)
- Week 4: create an EDA(Exploratory Data Analysis) and do QA (quality assurance) for the presentation (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions_V2/EDA.ipynb)
- Week 5: do more QA for the presentation
- Week 6: reflect on the pesentation

Weekly reflection: <https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/WeeklyReflections.md>

Learning goal II

Proficiency level: Undefined

I made an attempt to train a CNN using the DiverseDermatologyImages dataset which only had ~600 images, after many failed attempts I came to the conclusion that 600 images is simply not enough especially since they were divided into many classes some of which only had ~20 images.

Learning goal III

Proficiency level: Undefined

Evaluate Large Language Model concepts, focusing on their working, their applications and limitations, and their application in real-world cases based on the Data Science Methodology, taking ethical considerations into account.

Learning goal IV

Proficiency level: Orienting

Evaluate transfer learning principles and their importance for improving model performance with limited data, working on projects that use pre-trained models and datasets, leading to practical methods and positive feedback on their work.

Learning goal V

Proficiency level: Orienting

Week 1:

I missed the lesson due to a misunderstanding with the schedule but I contacted my group and tried to get up to speed with all the information I found on the course.

Week 2:

We worked in a group to discuss the previous week's research questions, I did not have everything researched myself so I tried to add to the conversation in other ways, like proposing questions or discussing vague terms.

Week 3:

I was assigned as the scribe this week so I mostly focused on writing down important notes from the meeting, however this week's discussion was pretty unorganized so it was hard to focus on both writing important notes as well as following the conversation.

(Document where I wrote the notes: https://github.com/FontysVenlo/groupepository-group-2/blob/main/ADAI_Skin_Cancer_Notes.md)

Week 4:

I was assigned QA for the presentation slides, however the day before the lesson the slides were still empty so I contacted the group to remind them to upload their information.